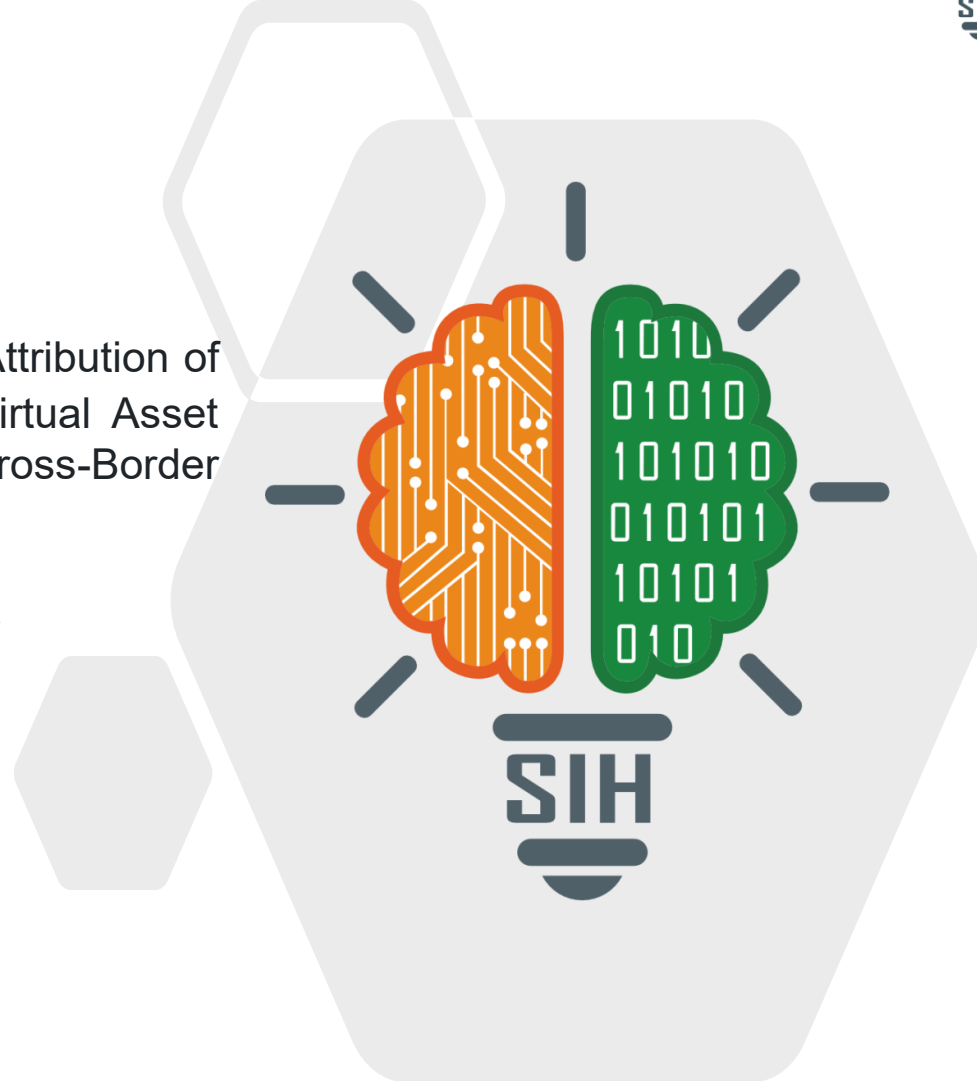


SMART INDIA HACKATHON 2026



- **Problem Statement ID** – 26182
- **Problem Statement Title-** Automated Attribution of Unknown Cryptocurrency Wallets to the Nearest Virtual Asset Service Providers (VASPs) for Stronger Cross-Border Investigations
- **Theme-** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-** 78
- **Team Name-** TROVER



PROPOSED SOLUTION

ANVESHAK (अन्वेषक, “the seeker”): An investigator console that turns an **unknown wallet** into the **nearest VASP holding the KYC**, shows **exactly why**, routes the **right legal request to the right country** and tracks it **until the money is frozen**.

- **Reads the complaint as typed** in Hinglish (“shaam 7:42”, “12.4 lakh”) and opens the case in seconds.
- **Remembers every wallet** India has ever reported.
- **Traces & attributes** across TRON, Ethereum & Bitcoin to the nearest VASP.
- **Names the crime type** — scam, ransomware, darknet or terror-financing signs.
- **Drafts & routes** BNSS/BSA notices, or MLAT / INTERPOL / FIU-to-FIU requests abroad.

KEY FEATURES

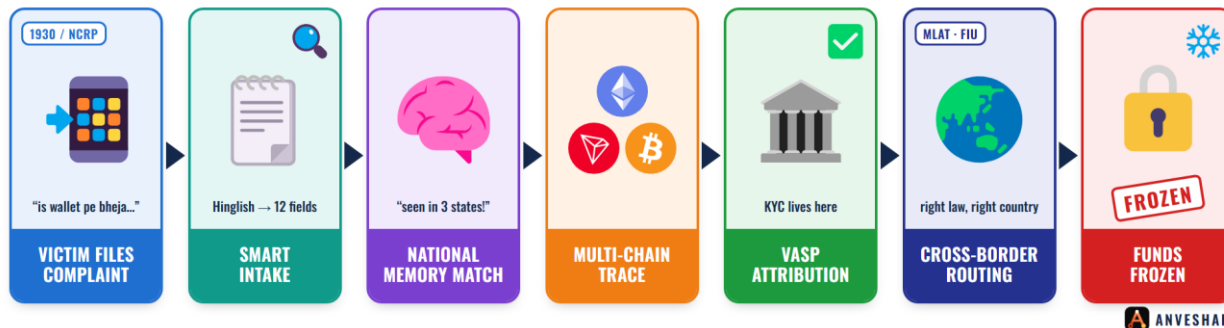
- ✓ **Smart Intake (complaint → case in seconds):** reads Hinglish, checksum-verifies every wallet, masks phones & UPI IDs at the door
- ✓ **SAHYOG National Memory (India never forgets a wallet):** the 2nd complaint from any state is matched instantly, with who saw it first
- ✓ **Golden-Hour Queue (work the case you can still save):** ranked by minutes left before the money reaches an exchange
- ✓ **Glass-Box Attribution:** every VASP verdict shows its evidence, weights and maths, plus “what would flip the verdict”

INNOVATION AND UNIQUENESS

- ✓ **Pre-emptive Freeze:** predicts the gang's next exchange and alerts it before the money lands.
- ✓ **Crime-Type Detection:** scam, ransomware, darknet or terror signs, with the maths shown.
- ✓ **Cold-Trail Re-acquisition:** operator habits re-find money lost in a mixer.
- ✓ **Cross-Border Autopilot:** foreign VASP? Auto-picks MLAT, INTERPOL or FIU-to-FIU.
- ✓ **Sanctions-Proximity Radar:** flags wallets “one shell away” from sanctions.
- ✓ **We Red-Team Ourselves:** 7 laundering tricks vs our own engine; misses published.

ANVESHAK — From Unknown Wallet to Frozen Funds

ONE COMPLAINT IN · NAMED EXCHANGE, ROUTED REQUEST AND A FREEZE OUT



WORKFLOW



Technologies to be used

Backend: Python, FastAPI, PostgreSQL, Redis, Celery

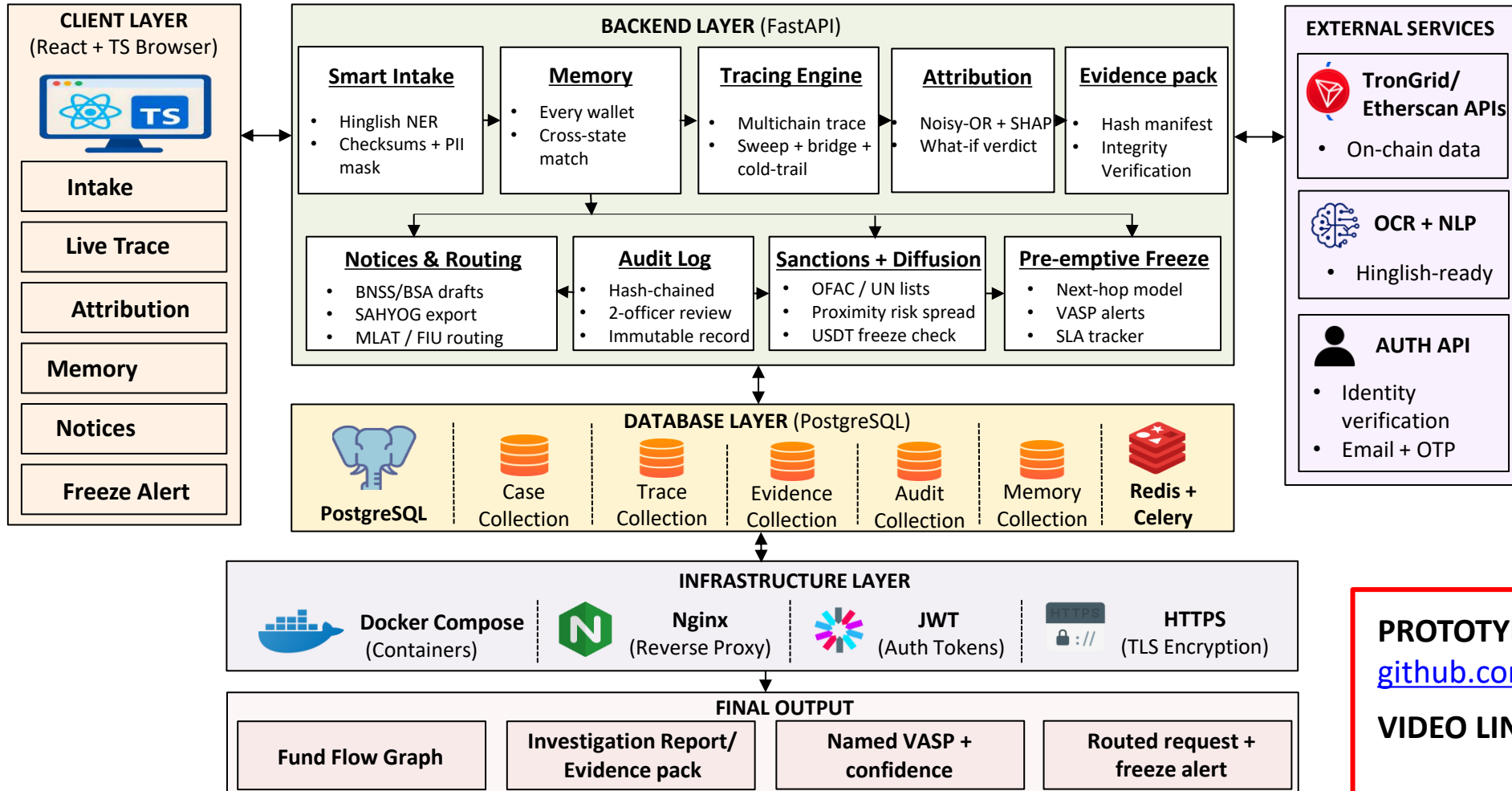
Graph & Detection: Rustworkx, Union-Find, BFS, risk diffusion

Machine Learning: LightGBM, SHAP, PyTorch Geometric

Frontend: React 19, TypeScript, Tailwind, Motion

Blockchain Data: TronGrid, Blockstream, Etherscan

Deployment: Docker Compose



PROTOTYPE LINK:

github.com/tripathidhruv/Anveshak

VIDEO LINK: to be added



FEASIBILITY:

Technical: Already running, not a concept: **490 passing tests**. Hinglish intake, checksum-verified wallets, national memory, multi-chain trace, risk and evidence pack, live on localhost today. Sweep signature needs **zero labelled** fraud data.

Operational: Single Docker host, **offline-capable**, no GPU. Imports NCRP read-only, exports **SAHYOG payloads**, so it slots into the I4C workflow police already use.

Social: Built for a constable, not a crypto analyst: reads complaints the way people actually write them. Every **score shows its factors**, so it can clear an innocent wallet, **not just accuse**.

Economical: Zero licensing vs. Chainalysis-class tools; one trace closes many cases, so cost per case falls as volume grows. **No recurring cloud/GPU cost**.

BUSINESS POTENTIAL:

State Licences:
Per-state licence for cyber cells: ₹25-50L/year
× 36 states & UTs = ₹9-18Cr/year at 70% margins.

VASP Compliance Feed:
FIU-registered exchanges subscribe to flagged-wallet alerts + SLA dashboard: 50 VASPs × ₹10-25L = ₹5-12Cr/year.

Global FIU Network: cross-border routing + Travel Rule checks as a service for FIUs across South & South-East Asia. By Year 5, ₹100Cr+ revenue.



VIABILITY:

People will use it:

- I4C gets one national graph, not 28 state silos
- An IO gets the exchange's name before the chai gets cold

Works at scale:

- Every complaint makes the memory smarter, no labelled data needed
- One trace closes many cases; syndicates surface on their own

Low risk to adopt:

- Plugs into NCRP and SAHYOG, no rip-and-replace
- Officer approves every output, no new legal liability

Potential Challenges and Strategies to overcome it

RISK	CHALLENGES	HOW WE OVERCOME IT
<u>Wrong VASP named</u>	Wrong exchange gets a notice	Glass-box verdict: every weight shown; infrastructure, never people; under 0.70 → human
<u>Trail dies in a mixer</u>	Most tools stop dead at the mixer	Cold-trail leads ranked by operator habits, always marked “lead, not proof”
<u>Offshore VASP</u>	Sits outside Indian law	Cross-border autopilot: MLAT / FIU-to-FIU, preserve-first request, SLA escalation

Potential impact on the target audience

Investigating Officer

Pastes a Hinglish complaint, gets the exchange's name. Zero crypto training needed.

I4C, SAHYOG and State Cells

One gang across 11 states shows up as one syndicate with one lead state, not 38 separate FIRs.

The Victim

The alert can be waiting at the exchange before the money lands. A real path to recovery replaces “nothing will happen.”

VASP Compliance Teams

Get structured, evidence-backed requests with a clear deadline and a public scorecard, so they act fast and refuse fewer.

Benefits of the solution



Social

- protects the elderly, students and first-time investors
- works in Hindi & Hinglish



Economic

- frozen before cash-out
- weeks of tracing cut
- ₹0 licence cost against lakhs per seat annually



Legal

- glass-box verdicts a defence lawyer can examine
- hash-chained, tamper-evident evidence



Strategic

- case data never leaves India
- national memory becomes a sovereign asset

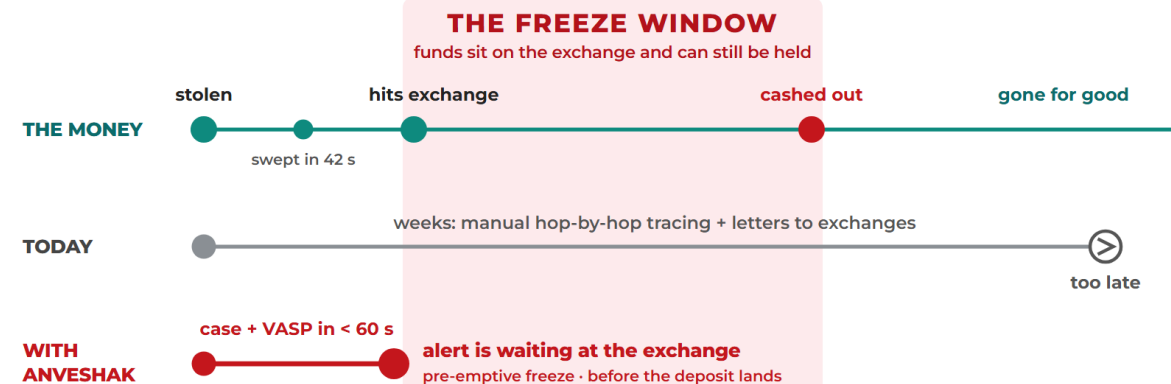


Environmental

- marginal, and stated honestly: no mining, no node syncing, commodity hardware

We race the money

A freeze works only while funds sit on the exchange. ANVESHAK gets there before the money lands.



timings illustrative

DEPLOYED LINK: to be added

VIDEO LINK: to be added

PROTOTYPE LINK:

<https://github.com/tripathidhruv/Anveshak>

Research & Benchmarking of Existing Systems

EXISTING PLATFORM	WHAT THEY DON'T HAVE	WHAT ANVESHAK HAS
Commercial Analytics Chainalysis · TRM Labs · Elliptic · Merkle Science	✗ Free access ✗ Open method ✗ Data stays in India	✓ Free, self-hostable ✓ Every weight shown ✓ Data never leaves India
Open-Source Analytics GraphSense · BlockSci	✗ Complaint intake ✗ National wallet memory ✗ TRON/USDT support	✓ Single-host Docker, OFFLINE ✓ Remembers every wallet ✓ TRON/USDT-first
Public Block Explorers Tronscan · Etherscan · Blockchair	✗ Wallet clustering ✗ Names receiving exchange ✗ Usable without expertise	✓ Automated multi-hop trace ✓ Names VASP + confidence ✓ No expert skill needed
NCRP / SAHYOG (I4C)	✗ Links cases across states ✗ Cross-border routing	✓ National memory + cross-border routing
Exchange / VASP Internal AML	✗ Acts before deposit ✗ Evidence-backed format	✓ Pre-emptive alert + evidence-backed request

References

Research & practice

- Weber, M. et al. "Anti-Money Laundering in Bitcoin." KDD Workshop, 2019 ([LINK](#))
- Möser, M. et al. "Money Laundering Tools in the Bitcoin Ecosystem." APWG eCrime, 2013 ([LINK](#))
- FATF — Risk-Based Approach to Virtual Assets and VASPs, 2021 ([LINK](#))

Datasets & label sources

- **Elliptic Data Set / Elliptic++** — labelled Bitcoin transactions
- **OFAC SDN crypto address list** — home.treasury.gov
- **BigQuery public blockchain datasets** — Bitcoin, Ethereum

Existing systems

- **NCRP** — ([LINK](#))
- **GraphSense** — ([LINK](#))
- **Chainalysis** — ([LINK](#))
- **Tronscan** — ([LINK](#))
- **TRM Labs** — ([LINK](#))
- **Etherscan** — ([LINK](#))